

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416491
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Buckley; John C. et al.

Method and system for measurement of coke drum deformation

Abstract

A coke drum measurement system includes a first target coupled to an upper aspect of a coke drum and a second target arranged generally perpendicular to the first target. A first optical measurement device is arranged to visualize the first target and measure a distance between the first optical measurement device and the first target. A second optical measurement device is arranged to visualize the second target and measure a distance between the second optical measurement device and the second target. A processor is electrically coupled to the first optical measurement device and the second optical measurement device. The processor receives signals from the first optical measurement device and the second optical measurement device corresponding to movement of the coke drum. The processor aggregates signals received from the first optical measurement device and the second optical measurement device to determine a total movement of the coke drum.

Inventors:	Buckley; John C. (Spring, TX), Clark; Robert J. (Kechi, KS)
Applicant:	AMEC FOSTER WHEELER USA CORPORATION (Houston, TX)
Family ID:	1000008820956
Assignee:	Amec Foster Wheeler USA Corporation (Houston, TX)
Appl. No.:	17/500250
Filed:	October 13, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220113128 A1	Apr. 14, 2022

Related U.S. Application Data

us-provisional-application US 63091602 20201014

Publication Classification

Int. Cl.: G01B11/16 (20060101); G01S7/48 (20060101); G01S17/42 (20060101)

U.S. Cl.:

CPC G01B11/16 (20130101); G01S7/4808 (20130101); G01S17/42 (20130101);

Field of Classification Search

CPC: G01B (11/16); G01B (11/026); G01S (17/42); G01S (7/4808)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5425279	12/1994	Clark et al.	N/A	N/A
9940702	12/2017	Clark et al.	N/A	N/A
2003/0066638	12/2002	Qu et al.	N/A	N/A
2010/0007729	12/2009	Clark et al.	N/A	N/A
2010/0326637	12/2009	Sasaki	118/695	H01L 21/67109
2012/0250004	12/2011	Parker	356/614	G01C 15/002
2014/0144071	12/2013	Dohi et al.	N/A	N/A
2014/0262731	12/2013	Meyer	N/A	N/A
2016/0274001	12/2015	Parker	N/A	G01C 15/002
2017/0145319	12/2016	Vivas Chacón	N/A	C10G 9/005
2019/0145769	12/2018	Sasaki	356/4.01	G01C 3/08
2019/0347783	12/2018	Salgian	N/A	G02B 27/017
2022/0213864	12/2021	Abe	N/A	F03B 15/08

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103868465	12/2013	CN	N/A
109959343	12/2018	CN	N/A

Primary Examiner: Ayub; Hina F

Attorney, Agent or Firm: WINSTEAD PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to, and incorporates by reference the entire disclosure of, U.S. Provisional Patent Application No. 63/091,602, filed on Oct. 14, 2020.

TECHNICAL FIELD

(1) The present application relates generally to coking systems and more particularly, but not by way of limitation, to systems and methods for measuring the deformation of a coke drum due to thermal expansion.

BACKGROUND

(2) This section provides background information to facilitate a better understanding of the various aspects of the disclosure. It should be understood that the statements in this section of this document are to be read in this light, and not as admissions of prior art.

(3) Coke-drum systems are frequently utilized in production of petroleum products such as, for example, gasoline, diesel fuel, fuel oil, and other similar products. A coke drum system typically includes a support pad or table top. A support structure is constructed above the support pad and a coke drum is disposed within the support structure. The coke drum is laterally unrestrained except at its support base ring and is vertically supported by the support pad. The coke drum is typically an enclosed metallic vessel often weighing on the order of several hundred tons. During operation, fluids enter the coke drum at high temperatures and induce thermal expansion of the coke drum. Further, water is typically introduced to the coke drum during operation thereby causing rapid cooling of the coke-drum contents. The introduction of heated fluids to the coke drum, and subsequent rapid cooling, induces significant temperature distributions inside the coke drum. Movement of the fluids within the coke drum often causes the temperature distributions to be uneven on the surface of the coke drum. Such uneven temperature distributions can lead non-uniform thermal expansion of the coke drum. During operation, it is common for the coke drum to bend to one side and assume a curved, banana-like shape. This phenomenon is commonly referred to as “the banana effect” or “banana movement”. Due to the size of the coke drum, such thermal expansion is often in the range of 4-6 inches. For this reason, there is minimal structural interconnection between the coke drum and the support structure so as to allow room for thermal expansion and contraction of the coke drum without damaging the coke drum or a surrounding support structure.

(4) While coke drums and their associated equipment are often designed to accommodate thermal expansion and banana movement, frequent inspection and monitoring is required to ensure that any movement of the coke drum remains within acceptable tolerances. Currently, visual inspection of the coke drum is utilized to monitor the condition of the coke drum and associated equipment and piping. However, visual inspection is time consuming and does not offer continuous monitoring of coke-drum movement. Excessive thermal expansion or banana movement can lead to material fatigue and cause premature failure of the coke drum and associated equipment and piping. Coke-drum equipment and piping failures result in production interruptions and considerable expense associated with repairs. Such failures can result in fire and potential injury to personnel.

SUMMARY

(5) This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of claimed subject matter.

(6) Aspects of the invention relate to a coke drum measurement system. The coke drum measurement system includes a first target coupled to an upper aspect of a coke drum and a second target coupled to the same or another upper aspect of the coke drum. The second target is arranged generally perpendicular to the first target. A first optical measurement device is arranged to visualize the first target and measure a distance between the first optical measurement device and the first target. A second optical measurement device is arranged to visualize the second target and measure a distance between the second optical measurement device and the second target. A processor is electrically coupled to the first optical measurement device and the second optical measurement device. The processor receives signals from the first optical measurement device and the second optical measurement device corresponding to movement of the coke drum. The processor aggregates signals received from the first optical measurement device and the second optical measurement device to determine a direction and a total movement of the coke drum.

(7) Aspects of the invention relate to a method of measuring coke drum deformation. The method

includes measuring a distance between a first target and a first optical measurement device along a first horizontal axis. The method also includes measuring a distance between a second target and a second optical measurement device along a second horizontal axis. A signal is transmitted from the first optical measurement device to a processor. The signal corresponds to movement of the coke drum along the first horizontal axis. A signal is transmitted from the second optical measurement device to the processor. The signal corresponds to movement of the coke drum along the second horizontal axis. A processor calculates a direction and a total movement of the coke drum.

(8) Aspects of the disclosure relate to a method of measuring movement of a coke drum. The method includes transmitting an electromagnetic beam from an optical measurement device to a target. A location of the electromagnetic beam to the target is recorded. Movement of the coke drum is determined based on movement of the electromagnetic beam to the target.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a schematic diagram of a coke drum system according to aspects of the disclosure;

(3) FIG. 2 is a schematic diagram of a two-axis coke-drum measurement system according to aspects of the disclosure;

(4) FIG. 3 is a plan view of a measurement target for use with the two-axis coke-drum measurement system of FIG. 2;

(5) FIG. 4A is a flow diagram illustrating a method of measuring coke-drum deformation according to aspects of the disclosure;

(6) FIG. 4B is a top view of the coke drum system illustrating an initial position of the coke drum;

(7) FIG. 4C is a top view of the coke drum system illustrating deflection of the coke drum along the first horizontal axis;

(8) FIG. 4D is a top view of the coke drum system illustrating deflection of the coke drum along the second horizontal axis;

(9) FIG. 4E is a top view of the coke drum system illustrating deflection of the coke drum along both the first horizontal axis and the second horizontal axis;

(10) FIG. 5 is a schematic diagram of a single-axis coke-drum measurement system according to aspects of the disclosure; and

(11) FIG. 6 is a flow diagram illustrating a method of measuring coke-drum deformation according to aspects of the disclosure.

DETAILED DESCRIPTION

(12) Various embodiments of the present invention will now be described more fully with reference to the accompanying drawings. The invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein.

(13) FIG. 1 is a schematic diagram of a coke drum system **100**. The coke drum system **100** includes a coke drum module **102** placed upon a support pad **104**. The coke drum module **102** includes a coke drum **106** placed inside a support frame **108**. By way of example, the coke drum system **100** illustrated in FIG. 1 shows four coke drum modules **102** and four support pads **104**. However, in other embodiments, coke drum systems utilizing principles of the invention may include any number of coke drum modules and any number of support pads.

(14) FIG. 2 is a schematic diagram of a two-axis coke-drum measurement system **200**. The two-axis coke-drum measurement system **200** includes a first target **202** associated with the coke drum

106 and a second target **204** associated with the coke drum **106**. The first target **202** is aligned with a first horizontal axis **206** of the coke drum **106** and the second target **204** is aligned with a second horizontal axis **208** of the coke drum **106** such that the second target **204** is arranged generally orthogonal to the first target **202**. As shown by way of example in FIG. 2, the first horizontal axis **206** is the “x” axis of the coke drum **106**, when the coke drum **106** is viewed from above and the second horizontal axis **208** is the “y” axis of the coke drum **106**. In various embodiments, the first target **202** and the second target **204** are coupled, for example, to an upper aspect of the coke drum **106**. In other embodiments, however, the first target **202** and the second target **204** may be coupled to any exterior portion of the coke drum **106**. In still other embodiments, the first target **202** and the second target **204** may be coupled to other components of the coke drum system **100** that move during thermal expansion and contraction of the coke drum **106** such as, for example, components of the piping or devices that move during thermal expansion and contraction of the coke drum **106**.

(15) Still referring to FIG. 2, the two-axis coke-drum measurement system **200** includes a first optical measurement device **210** and a second optical measurement device **212**. The first optical measurement device **210** is arranged to visualize the first target **202** and the second optical measurement device **212** is arranged to visualize the second target **204**. In a typical embodiment, the first optical measurement device **210** and the second optical measurement device **212** are, for example, laser measurement devices; however, other devices could be utilized including, for example, infra-red devices, electromagnetic devices, or other devices capable of measuring a distance without contact with an accuracy of, for example, approximately ± 3 mm. In various embodiments, the first optical measurement device **210** and the second optical measurement device **212** could be, for example, a Micro-Epsilon model ILR2250-100 meter available from Micro-Epsilon. The first optical measurement device **210** measures a linear distance between the first target **202** and the first optical measurement device **210** along the first horizontal axis **206**. The second optical measurement device **212** measures a linear distance between the second target **204** and the second optical measurement device **212** along the second horizontal axis **208**. In various embodiments, the first optical measurement device **210** and the second optical measurement device **212** measure distances utilizing any optical-distance-measurement methodology including, for example, time-of-flight measurement, triangulation, vision-based measurement, confocal sensing, interferometry, conoscopic holography, or other appropriate method. In various embodiments, the first optical measurement device **210** includes a first light source **214** and a first measurement unit **216**. In various embodiments, the second optical measurement device **212** includes a second light source **218** and a second measurement unit **220**. In various embodiments, the first light source **214** and the first measurement unit could be separate or integral devices. Similarly, the second light source **218** and the second measurement device **220** could be separate or integral devices.

(16) Still referring to FIG. 2, the first optical measurement device **210** and the second optical measurement device **212** are coupled to a processor **222** via a data bus **215**. In various embodiments, the data bus **215** could be a wired or a wireless coupling. In a various embodiments, the data bus **215** may include, for example, any combination of hardware, software embedded in a computer readable medium, or encoded logic incorporated in hardware or otherwise stored (e.g., firmware) to couple components of the two-axis coke-drum measurement system **200** to each other. As an example and not by way of limitation, the data bus **215** may include an Accelerated Graphics Port (AGP) or other graphics bus, a Controller Area Network (CAN) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCI-X) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or any other suitable bus or a combination of two or more of these. In various embodiments, the data bus **215** may include any number, type, or configuration of data buses **215**, where appropriate.

(17) The processor **222** may be any microprocessor, microcontroller, programmable element, or

other device or collection of devices for aggregating and computing measurements provided by the first optical measurement device **210** and the second optical measurement device **212**. The processor **222** receives, for example, signals from the first optical measurement device **210** corresponding to changes in distance between the first target **202** and the first optical measurement device **210** along the first horizontal axis **206**. The processor **222** also receives, for example, signals from the second optical measurement device **212** corresponding to changes in distance between the second target **204** and the second optical measurement device **212** along the second horizontal axis **208**. During operation the processor **222** aggregates the measurements received from the first optical measurement device **210** and the second optical measurement device **212** in order to determine an aggregate movement of the coke drum **106**. In various embodiments, the processor **222** is coupled to an output device **224**. The output device provides real-time measurements of deformation of the coke drum **106** and provides an alert when deformation of the coke drum **106** exceeds a pre-determined threshold. In various embodiments, the pre-determined threshold may be, for example, approximately 2 inches in any direction. In other embodiments, any threshold could be utilized. In various embodiments, the alert may be, for example, an auditory alert, a visual alert, or a combination of an auditory alert and a visual alert. In various embodiments, the processor **222** may initiate corrective action when the deformation of the coke drum **106** exceeds the pre-determined threshold. For example, in various embodiments, the processor **222** may direct a reduction in the rate that water is added to the coke drum **106** when the coke drum **106** is being cooled. Such a reduction in the addition of water reduces a rate of cooling of the coke drum **106** and lessens thermal deformation of the coke drum **106**.

(18) FIG. 3 is a plan view of a measurement target **300** for use with the two-axis coke-drum measurement system **200**. For purposes of discussion, the measurement target **300** is representative of both the first target **202** and the second target **204**. In various embodiments, the measurement target **300** is rectangular; however, in other embodiments, the measurement target **300** could be any appropriate shape. The measurement target **300** includes a vertical dimension **302**. The vertical dimension **302** has a length that is approximately equal to the anticipated vertical thermal growth of the coke drum **106** plus, for example, 50 mm. Such a dimension allows a measurement device such as the first optical measurement device **210** or the second optical measurement device **212** to visualize the measurement target **300** as the measurement target **300** moves with the vertical thermal expansion of the coke drum **106**. The measurement target **300** includes a horizontal dimension **304**. The horizontal dimension **304** has a length that is approximately equal to two times the theoretical anticipated banana movement of the coke drum **106**. Such a dimension allows a measurement device such as the first optical measurement device **210** or the second optical measurement device **212** to visualize the measurement target **300** as the coke drum **106** experiences lateral banana movement.

(19) Referring to FIGS. 1-3 collectively, during operation, the first optical measurement device **210** initially focuses light along the first horizontal axis **206** (i.e. the “x” direction) on the top center of the first target **202**. As the coke drum **106** undergoes thermal expansion in the vertical direction (i.e. the “z” direction), the first target **202** moves upwardly relative to the first optical measurement device **210**. Thus, the light focused by the first optical measurement device **210** becomes more centered top to bottom on the first target **202**. As the coke drum **106** experiences banana movement along the second horizontal axis **208** (i.e. the “y” direction), the light focused by the first optical measurement device **210** moves to the right or to the left on the first target **202**. Additionally, as the coke drum **106** experiences banana movement along the second horizontal axis **208** (i.e. the “y” direction), the distance between the second target **204** and the second optical measurement device **212** may increase or decrease.

(20) Still referring to FIGS. 1-3 collectively, during operation, the second optical measurement device **212** initially focuses light along the second horizontal axis **208** (i.e. the “y” direction) on the top center of the second target **204**. As the coke drum **106** undergoes thermal expansion in the

vertical direction (i.e. the “z” direction), the second target **204** moves upwardly relative to the second optical measurement device **212**. Thus, the light focused by the second optical measurement device **212** becomes more centered top to bottom on the second target **204**. As the coke drum **106** experiences banana movement along the first horizontal axis **206** (i.e. the “x” direction), the light focused by the second optical measurement device **212** moves to the right or to the left on the second target **204**. Additionally, as the coke drum **106** experiences banana movement along the first horizontal axis **206** (i.e. the “x” direction), the distance between the first target **204** and the first optical measurement device **212** may increase or decrease.

(21) FIG. 4A is a flow diagram illustrating a process **400** of measuring deformation of the coke drum **106**. The process **400** begins at step **402**. At step **404**, the first optical measurement device **210** focuses light on the first target **202**. At step **405**, the second optical measurement device **212** focuses light on the second target **204**. At step **406**, the first optical measurement device **210** measures and transmits to the processor **222** a first initial linear distance between the first target **202** and the first optical measurement device **210** along the first horizontal axis **206**. In various embodiments, the first initial linear distance between the first target **202** and the first optical measurement device **210** is measured prior to operation of the coke drum **106** and may, for example, be defined as “0” by the processor **222**. At step **407**, the second optical measurement device **212** measures and transmits to the processor **222**, a second initial linear distance between the second target **204** and the second optical measurement device **212** along the second horizontal axis **208**. In various embodiments, the second initial linear distance between the second target **204** and the second optical measurement device **212** is measured prior to operation of the coke drum **106** and may, for example, be defined as “0” by the processor **222**. At step **408**, the first optical measurement device **210** measures and transmits to the processor **222** a signal corresponding to a first operational distance along the first horizontal axis **206** between the first target **202** and the first optical measurement device **210**. In various embodiments, the first operational distance is measured during operation of the coke drum **106**. At step **409**, the second optical measurement device **212** measures and transmits to the processor **222** a signal corresponding to a second operational distance along the second horizontal axis **208** between the second target **204** and the second optical measurement device **212**. In various embodiments, the second operational distance is measured during operation of the coke drum **106**. At step **410**, the processor **222** determines a change in distance along the first horizontal axis **206** and a change in distance along the second horizontal axis **208**. The processor **222** aggregates the signals from the first optical measurement device **210** and the second optical measurement device **212** in order to determine a total movement of the coke drum **106**.

(22) Still referring to FIG. 4A, at step **412**, the processor transmits the total movement of the coke drum **106** to the output device **224**. At step **414**, the processor **222** compares the total movement of the coke drum **106** to a pre-determined alert threshold and determines if the total movement of the coke drum **106** exceeds the pre-determined alert threshold. If, at step **414**, it is determined that the total movement of the coke drum **106** does not exceed the pre-determined alert threshold, the process **400** returns to step **402**. If at step **414**, it is determined that the total movement of the coke drum **106** exceeds the pre-determined alert threshold, then the processor proceeds to step **416**, where a visual or auditory alert is generated. In various embodiments, the alert prompts intervention by an operator of the coke drum **106**. In various embodiments, the intervention may include, for example, slowing a rate that water is added to the coke drum in an effort to reduce the cooling rate of the coke drum **106**. At step **418**, the processor **222** determines if movement of the coke drum **106** exceeds a pre-determined mitigation threshold. If, at step **418**, it is determined that movement of the coke drum **106** exceeds the pre-determined mitigation threshold, the process **400** proceeds to step **420**. At step **420**, the processor **222** may, in various embodiments, direct automatic correction. In various embodiments, the corrective action may include the processor **222** automatically slowing a rate that water is added to the coke drum in an effort to reduce the cooling

rate of the coke drum **106**. The process **400** then returns to step **402**. Thus, the two-axis coke-drum measurement system **200** is capable of providing continuous real-time measurement of deformation of the coke drum **106**.

(23) FIG. **4B** is a top view of the coke drum system **100** illustrating an initial position of the coke drum **106**. FIG. **4C** is a top view of the coke drum system **100** illustrating deflection of the coke drum **106** along the first horizontal axis **206**. FIG. **4D** is a top view of the coke drum system **100** illustrating deflection of the coke drum **106** along the second horizontal axis **208**. FIG. **4E** is a top view of the coke drum system **100** illustrating deflection of the coke drum **106** along both the first horizontal axis **206** and the second horizontal axis **208**. Referring to FIGS. **4B-4E** collectively, the first optical measurement device **210** measures a change in distance between the first target **202** and the first optical measurement device **210** along the first horizontal axis **206**. The change in distance between the first target **202** and the first optical measurement device **210** is represented graphically by deflected first target **202'**. The second optical measurement device **212** measures a change in distance between the second target **204** and the second optical measurement device **212** along the second horizontal axis **208**. The change in distance between the second target **204** and the second optical measurement device **212** is represented graphically by deflected second target **204'**. The processor **222** then aggregates the signals received from the first optical measurement device **210** and the second optical measurement device **212**, for example, according to the Pythagorean theorem, to determine a total movement of the coke drum **106**.

(24) FIG. **5** is a schematic diagram of a single-axis coke-drum measurement system **500**. The single-axis coke-drum measurement system **500** includes an optical measurement device **502** positioned near an upper aspect of the coke drum **106**. In various embodiments, the optical measurement device **502** is coupled to an outer surface of the coke drum **106**; however, in other embodiments, the optical measurement device **502** may be coupled to other components of the coke drum system **100** that move with the thermal expansion of the coke drum **106** and may not be directly coupled to the coke drum **106**. In various embodiments, the optical measurement device **502** is, for example, a laser measurement device; however, other devices could be utilized including, for example, an infra-red device, an electromagnetic device, or other devices capable of directing an electromagnetic beam **508** along a vertical length of the coke drum **106** to a target **506** that is positioned near a base of the coke drum **106**. The target **506** is coupled to components of the coke drum system **100** that do not move with the thermal expansion of the coke drum **106** such as, for example, the support pad **104**. In various embodiments, the target **506** is able to detect movement of the electromagnetic beam **508** in both an “x” and a “y” direction thereby detecting banana movement of the coke drum **106**. The target **506** is electrically coupled to the processor **222**. During operation, the target **506** provides a signal to the processor **222** corresponding to a position of the electromagnetic beam **508** on the target **506**. The processor **222** converts the signal to a measurement of movement of the coke drum **106** and provides an alert if movement of the coke drum **106** exceeds a pre-determined threshold. In various embodiments, the alert could be, for example, a visual alert, an auditory alert, or a combination of a visual alert and an auditory alert. The pre-determined threshold could be, for example, ± 5 mm.

(25) FIG. **6** is a flow diagram illustrating a process **600** of measuring coke-drum deformation. The process **600** begins at step **602**. At step **604**, the optical measurement device **502** transmits an electromagnetic beam **508** to the target **506**. At step **606**, the target records a position of the electromagnetic beam **508** on the target **506** and transmits a signal to the processor **222** corresponding to a position of the electromagnetic beam **508** on the target **506**. At step **608**, the target **506** records movement of the electromagnetic beam **508** on the target **506** and transmits a signal to the processor **222** corresponding to a location of the electromagnetic beam **508** on the target **506**. At step **610**, the processor **222** interprets the signals received from the target **506** to determine a movement of the coke drum **106**. At step **612**, the processor determines if a degree of movement of the coke drum **106** exceeds a pre-determined alert threshold. If, at step **612**, it is

determined that the movement of the coke drum **106** does not exceed the pre-determined alert threshold, the process **600** returns to step **604**. If, at step **612**, it is determined that the movement of the coke drum **106** exceeds the pre-determined alert threshold, the process **600** proceeds to step **614**, where an auditory or visual alert is generated. In various embodiments, the alert prompts intervention by an operator of the coke drum **106**. In various embodiments, the intervention may include, for example, slowing a rate that water is added to the coke drum in an effort to reduce the cooling rate of the coke drum **106**. At step **616**, the processor **222** determines if movement of the coke drum **106** exceeds a pre-determined mitigation threshold. If, at step **616**, it is determined that movement of the coke drum **106** exceeds the pre-determined mitigation threshold, the process **600** proceeds to step **618**. At step **618**, the processor **222** may, in various embodiments, direct automatic correction. In various embodiments, the corrective action may include the processor **222** automatically slowing a rate that water is added to the coke drum in an effort to reduce the cooling rate of the coke drum **106**. The process **600** then returns to step **604**. Thus, the single-axis coke-drum measurement system **500** is capable of providing continuous real-time measurement of deformation of the coke drum **106**.

(26) The term “substantially” is defined as largely but not necessarily wholly what is specified, as understood by a person of ordinary skill in the art. In any disclosed embodiment, the terms “substantially,” “approximately,” “generally,” and “about” may be substituted with “within [a percentage] of” what is specified, where the percentage includes 0.1, 1, 5, and 10 percent.

(27) The term “at least one of” is meant to cover combinations of the listed elements, components, features, and the like, as well as the listed elements, components, features, and the like individually. For example, the phrase “at least one of A and B” is meant to cover embodiments comprising only A, embodiments comprising only B, and embodiments comprising both A and B unless stated otherwise.

(28) Conditional language used herein, such as, among others, “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

(29) While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, the processes described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. The scope of protection is defined by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A coke drum measurement system comprising: a first target coupled to an upper aspect of a coke drum; a second target coupled to the upper aspect of the coke drum and arranged generally perpendicular to the first target; a first optical measurement device that visualizes the first target and measures a distance between the first optical measurement device and the first target; a second optical measurement device that visualizes the second target and measures a distance between the second optical measurement device and the second target; a processor electrically coupled to the first optical measurement device and the second optical measurement device, the processor

receiving signals from the first optical measurement device and the second optical measurement device corresponding to movement of the coke drum, wherein the processor aggregates signals received from the first optical measurement device and the second optical measurement device to determine a total movement of the coke drum; wherein the processor is configured to generate an alert in response to a determination that the total movement exceeds a pre-determined alert threshold; and wherein the processor is configured to initiate a corrective action in response to a determination that the total movement exceeds a pre-determined mitigation threshold.

2. The coke drum measurement system of claim 1, wherein the first optical measurement device measures a distance between the first optical measurement device and the first target along a first horizontal axis.

3. The coke drum measurement system of claim 2, wherein the second optical measurement device measures a distance between the second optical measurement device and the second target along a second horizontal axis that is generally perpendicular to the first horizontal axis.

4. The coke drum measurement system of claim 1, wherein the first optical measurement device and the second optical measurement device are laser measurement devices.

5. The coke drum measurement system of claim 1, wherein the first target comprises a vertical dimension generally equal to an anticipated vertical thermal growth of the coke drum plus 50 mm.

6. The coke drum measurement system of claim 5, wherein the first target comprises a horizontal dimension generally equal to two times an anticipated banana movement of the coke drum.

7. The coke drum measurement system of claim 1, wherein the second target comprises a vertical dimension generally equal to an anticipated vertical thermal growth of the coke drum plus 50 mm.

8. The coke drum measurement system of claim 7, wherein the second target comprises a horizontal dimension generally equal to two times an anticipated banana movement of the coke drum.

9. A method of measuring coke drum deformation, the method comprising: measuring a distance between a first target and a first optical measurement device along a first horizontal axis; measuring a distance between a second target and a second optical measurement device along a second horizontal axis; transmitting a signal from the first optical measurement device to a processor, the signal corresponding to movement of the coke drum along the first horizontal axis; transmitting a signal from the second optical measurement device to the processor, the signal corresponding to movement of the coke drum along the second horizontal axis; calculating, via a processor a total movement of the coke drum, wherein the total movement is an aggregate of movement along the first and second horizontal axes; initiating, via the processor, an alert in response to a determination that the total movement exceeds a pre-determined alert threshold; and initiating, via the processor, a corrective action in response to a determination that the total movement exceeds a pre-determined mitigation threshold.

10. The method of claim 9, comprising directing an electromagnetic beam from the first optical measurement device to the first target.

11. The method of claim 9, comprising directing an electromagnetic beam from the second optical measurement device to the second target.

12. The method of claim 9, wherein the corrective action comprises reducing a rate at which water is added to the coke drum.

13. The method of claim 9, wherein the alert comprises a combination of a visual and an auditory alert.

14. The method of claim 13, wherein the alert is either an auditory or a visual alert.

15. A method of measuring movement of a coke drum, the method comprising: transmitting an electromagnetic beam from an optical measurement device to a target; recording a location of the electromagnetic beam on the target; and determining movement of the coke drum based on movement of the electromagnetic beam on the target; initiating, via the processor, an alert in response to a determination that the movement exceeds a pre-determined alert threshold; and

initiating a corrective action in response to a determination that the movement of the coke drum exceeds a pre-determined mitigation threshold, the corrective action comprising reducing a rate of cooling of the coke drum.

16. The method of claim 15, wherein the optical measurement device is disposed proximate an upper aspect of the coke drum.

17. The method of claim 16, wherein the optical measurement device moves with the coke drum as the coke drum deforms.

18. The method of claim 15, wherein the target is disposed proximate a lower aspect of the coke drum.

19. The method of claim 18, wherein the target does not move with the coke drum as the coke drum deforms.

20. The method of claim 15, wherein the alert comprises a combination of a visual and an auditory alert.
